

Mutual-visibility sets in Andrásfai graphs

A four-state proof of OEIS A391632, with cardinality refinements

ChatGPT

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Abstract

We prove the rational generating function conjectured by Andrew Howroyd in OEIS A391632 for the number of mutual-visibility sets in Andrásfai graphs. The proof translates the common-neighbor obstruction into an infinite family of forbidden cyclic binary patterns, then recognizes that family with a four-state weighted automaton. A uniqueness argument for periodic auxiliary flags is essential: it makes the trace of a matrix power count each labeled vertex subset exactly once. Keeping a variable for cardinality gives the full bivariate generating function. We also obtain an explicit binomial formula, classify the maximum sets, prove eventual polynomiality at every fixed deficit from maximum size, derive exponential growth and a central limit theorem, and give exact recognition and sampling algorithms. Independent exhaustive checks cover all 9,586,980 subsets of the first eight graphs; symbolic checks verify the displayed algebra. The proof is self-contained and does not depend on those finite checks.

Status and scope. The OEIS entry, consulted on 19 September 2026, still labels the generating function as conjectured [1]. The argument here establishes that statement and the refinements below. Targeted literature searches did not identify an earlier proof, but they do not certify global priority. This report has not been independently refereed or formalized in a proof assistant. The executable checks and all source files accompany the report.

Keywords. Mutual visibility; Andrásfai graph; cyclic word; finite automaton; transfer matrix; rational generating function; fixed-deficit enumeration.

Main identity. For the size-enumerator $V_n(y)$ of the n th graph,

$$V_n(y) = \operatorname{tr} \left(\begin{pmatrix} 1 & y & 0 & 0 \\ 1 & 0 & y & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1+y & 0 & 0 \end{pmatrix}^{3n-1} \right) \quad (n \geq 2).$$

The exceptional graph $\operatorname{And}(1) = K_2$ has $V_1(y) = (1+y)^2$.

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1 The conjecture and the enumeration problem

A set S of vertices of a graph is a *mutual-visibility set* when every pair of distinct vertices of S can be joined by a shortest path whose internal vertices lie outside S . The condition is existential: one suitable shortest path is enough. It does not require every shortest path to avoid S . We use the definition introduced by Di Stefano [2]. In particular, the empty set, every singleton, and every two-element set are mutual-visibility sets in any connected graph.

We use the standard circulant realization of the Andrásfai graph [3]. For an integer $n \geq 1$, put

$$N = 3n - 1, \quad G_n = \text{And}(n) = \text{Cay}(\mathbb{Z}/N\mathbb{Z}, \{1, 4, 7, \dots, 3n - 2\}). \quad (1)$$

All vertex indices below are taken modulo N . The connection set is closed under negation, since $N \equiv 2 \pmod{3}$; thus the graph is undirected. Each vertex has n neighbors. Our count is on this fixed labeled vertex set: rotations and other graph automorphisms do *not* identify subsets.

Write

$$\mathcal{M}_n = \{S \subseteq \mathbb{Z}/N\mathbb{Z} : S \text{ is a mutual-visibility set of } G_n\}, \quad V_n(y) = \sum_{S \in \mathcal{M}_n} y^{|S|}, \quad a_n = V_n(1). \quad (2)$$

The variable y marks the cardinality of the set; x will mark the graph index n . This distinction matters because the number of vertices is $3n - 1$, not n .

OEIS A391632 records a_n , beginning

$$4, 21, 127, 749, 4455, 26725, 161007, 971613, \dots$$

The entry was created by Eric W. Weisstein on 14 December 2025. On 12 January 2026, Andrew Howroyd supplied the following conjecture [1]:

$$\sum_{n \geq 1} a_n x^n = \frac{x(4 - 19x + 33x^2 - 40x^3 + 8x^4)}{1 - 10x + 29x^2 - 32x^3 + 8x^4}. \quad (3)$$

The goal is to explain this identity combinatorially, not simply to extend the sequence or fit its recurrence.

Theorem 1.1 (Counting theorem). *Let*

$$M(y) = \begin{pmatrix} 1 & y & 0 & 0 \\ 1 & 0 & y & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1+y & 0 & 0 \end{pmatrix}. \quad (4)$$

Then, for every $n \geq 1$,

$$V_n(y) = \text{tr } M(y)^{3n-1} + \mathbf{1}_{\{n=1\}} y^2. \quad (5)$$

Consequently, equation (3) holds.

The proof occupies sections 2 to 4. Its difficult point is not matrix inversion. It is proving that a cyclic word corresponds to exactly one closed automaton walk; an arbitrary deterministic automaton for linear words would not by itself justify a trace formula.

2 From shortest paths to forbidden cyclic patterns

2.1 The common-neighbor interval

Let $\Gamma(v)$ denote the neighborhood of a vertex v in G_n .

Lemma 2.1. *For $n \geq 2$, every unordered nonadjacent pair has a unique orientation $(a, a + 3r - 1)$ with $1 \leq r \leq n - 1$. Its common neighbors are exactly*

$$\Gamma(a) \cap \Gamma(a + 3r - 1) = \{a + 1, a + 4, \dots, a + 3r - 2\}. \quad (6)$$

In particular, G_n has diameter two.

Proof. For a nonedge, the two positive oriented differences sum to N and neither is congruent to 1 modulo 3. Since $N \equiv 2 \pmod{3}$, one difference is congruent to 0 and the other to 2. The latter is uniquely $3r - 1$ with $1 \leq r \leq n - 1$.

By translation, take $a = 0$ and write $b = 3r - 1$. A neighbor of 0 is a vertex $v = 1 + 3j$, with $0 \leq j \leq n - 1$. If $j \leq r - 1$, then $v < b$ and

$$b - v = 3(r - j) - 2 \equiv 1 \pmod{3},$$

so v is adjacent to b . If $j \geq r$, then $v > b$ and $v - b \equiv 2 \pmod{3}$, so it is not adjacent to b . This proves the displayed intersection. It is nonempty, so each nonedge has distance two; nonedges exist for $n \geq 2$. $\square$

Adjacent selected vertices are always visible to one another using their edge. For a nonadjacent pair, a shortest path has one internal vertex, and that vertex must be a common neighbor. Therefore visibility fails precisely when the endpoints and *all* their common neighbors belong to S .

2.2 Membership words and the infinite forbidden family

Encode S by its periodic membership word $(w_t)_{t \in \mathbb{Z}}$, with period N , where $w_t = 1$ exactly when $t \bmod N \in S$. By [theorem 2.1](#), S fails to be mutually visible precisely when there are a and r , with $1 \leq r \leq n - 1$, such that

$$w_a = w_{a+3r-1} = w_{a+1} = w_{a+4} = \dots = w_{a+3r-2} = 1. \quad (7)$$

If a star denotes an arbitrary binary symbol, this is the pattern

$$\mathcal{F}_r = 11(* * 1)^{r-1}1. \quad (8)$$

For example, the first three patterns are 111, 11**11, and 11**1**11. The stars are independent: they impose no requirement of being zero.

The original graph only supplies the range $r \leq n - 1$. For a uniform finite-state description, we need to remove that bound without introducing new restrictions.

Lemma 2.2 (Periodic extension). *For $n \geq 2$, a period- N word avoids $\mathcal{F}_r$ for $1 \leq r \leq n - 1$ if and only if it avoids $\mathcal{F}_r$ for every $r \geq 1$.*

Proof. Only the forward direction needs proof. Suppose an occurrence of $\mathcal{F}_r$ begins at a with $r \geq n$. Among its mandatory ones are w_a, w_{a+1} , and w_{a+3n-2} . Since $3n - 2 = N - 1$, the last of these is w_{a-1} by periodicity. Thus $w_{a-1}w_a w_{a+1} = 111$, an occurrence of $\mathcal{F}_1$. Since $n \geq 2$, that pattern was already forbidden. $\square$

It is important that the first graph is excluded here. For $n = 1$, $G_1 = K_2$ has no nonedges and all four subsets are valid. Repeating its full-set word 11 nevertheless produces 111. This is the source of the correction term in [equation \(5\)](#), and later of the precise starting index of the scalar recurrence.

3 A four-state automaton, with a cyclic uniqueness proof

3.1 Auxiliary flags retain arbitrarily long prefixes

For a binary word indexed by all integers, let b_t be true if a finite pattern of the form

$$11(*1)^{r-1}, \quad r \geq 1, \quad (9)$$

ends at position t . This is the forbidden pattern with its final 1 removed. The flag asks whether reading another 1 next would complete an obstruction.

Using Boolean multiplication for conjunction, its recurrence is

$$b_t = w_t(w_{t-1} \vee b_{t-3}). \quad (10)$$

Indeed, either the prefix is the initial 11, or an earlier such prefix ends three positions before t and is extended by two arbitrary bits and a final 1. The word avoids all forbidden patterns if and only if

$$b_t w_{t+1} = 0 \quad \text{for every } t. \quad (11)$$

An apparent problem remains: a recurrence on a circle can have several solutions, or a solution supported by a circular chain with no finite starting witness. The arithmetic of N rules this out.

Lemma 3.1 (Uniqueness of periodic flags). *Let $N = 3n - 1$ and let a period- N binary word contain a zero. There is exactly one period- N Boolean solution of [equation \(10\)](#). That solution has the finite-prefix interpretation [equation \(9\)](#).*

Proof. Choose z with $w_z = 0$. Necessarily $b_z = 0$. Since $\gcd(3, N) = 1$, the residues

$$z, z + 3, z + 6, \dots, z + 3(N - 1)$$

run through all vertices. Starting with $b_z = 0$, recurrence [equation \(10\)](#) determines the flag at each subsequent residue from the previous one. On returning to z , the equation still reads $b_z = 0$ regardless of the incoming flag. Thus a unique periodic solution exists.

The flags defined by finite occurrences of [equation \(9\)](#) are periodic and satisfy the same recurrence: a witnessed prefix either has length two or has a shorter witnessed prefix three positions earlier. Conversely, both terms on the right of [equation \(10\)](#) create a finite witness. Hence those semantic flags form a solution, and uniqueness identifies it with the one just constructed. Equivalently, backwards substitution by steps of three must encounter a zero, preventing an unsupported circular chain. $\square$

The all-ones word is never accepted: [equation \(10\)](#) forces all flags to one, which violates [equation \(11\)](#). Consequently, the uniqueness lemma applies to every accepted word.

3.2 Six semantic states

Consider the state immediately after reading position t :

$$(w_t, b_t, b_{t-1}, b_{t-2}).$$

Under equations (10) and (11), no two of b_t, b_{t-1}, b_{t-2} can be one. Adjacent flags are impossible because $b_{t-1} = 1$ forces $w_t = 0$, whereas $b_t = 1$ forces $w_t = 1$. If $b_t = b_{t-2} = 1$, then $w_{t-1} = 0$ by equation (11); equation (10) then forces $b_{t-3} = 1$, contradicting the impossibility of the adjacent pair $b_{t-3} = b_{t-2} = 1$.

The only possible states and their transitions are therefore the following. A dash means rejection. Edge labels are the bit being read, not a state weight.

State	$(w_t, b_t, b_{t-1}, b_{t-2})$	Read 0	Read 1
A	$(0, 0, 0, 0)$	A	B_1
B_1	$(1, 0, 0, 0)$	A	C
C	$(1, 1, 0, 0)$	D	—
D	$(0, 0, 1, 0)$	B_2	B_3
B_2	$(0, 0, 0, 1)$	A	C
B_3	$(1, 0, 0, 1)$	A	C

To check the table mechanically, from (a, b, c, d) a new bit ε is forbidden when $b\varepsilon = 1$. Otherwise the new state is

$$(a, b, c, d) \mapsto (\varepsilon, \varepsilon(a \vee d), b, c). \quad (12)$$

Thus the table is an exact local restatement of the flag recurrence and the rejection rule, not a guessed automaton.

3.3 Merging states without changing cyclic multiplicities

The three states B_1, B_2, B_3 have identical outgoing labeled transitions. Merge them to a single state B . The four-state labeled graph is shown in figure 1.

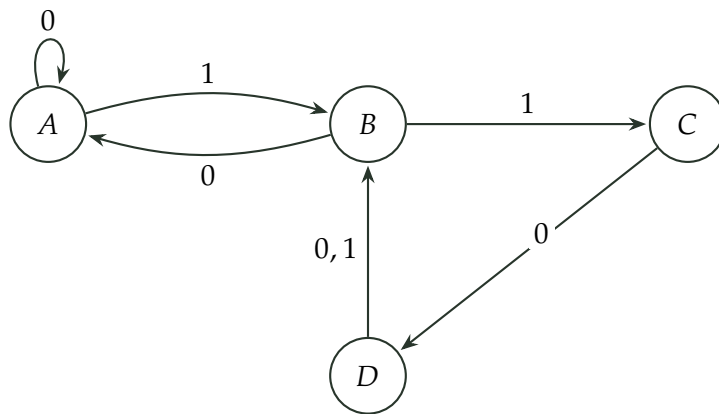


Figure 1: The four-state labeled automaton. The $D \rightarrow B$ arrow denotes two distinct edges, with labels 0 and 1. Weighting a label 1 by y and a label 0 by 1 gives the matrix $M(y)$.

A quotient of a state graph need not preserve the number of closed walks. Here it does, for an explicit reason. Each incoming edge to B determines the original state uniquely:

$$A \xrightarrow{1} B \rightsquigarrow B_1, \quad D \xrightarrow{0} B \rightsquigarrow B_2, \quad D \xrightarrow{1} B \rightsquigarrow B_3. \quad (13)$$

There are no other incoming edges to B . The other states lift identically. Every labeled closed walk in the quotient therefore lifts to exactly one labeled closed walk in the six-state graph, and conversely.

Proposition 3.2 (Anchored cyclic bijection). *For $n \geq 2$, the mutual-visibility sets of G_n are in weight-preserving bijection with the length- N closed labeled walks in the four-state graph, with a fixed time origin. The weight of a walk is y to the number of its edges labeled 1.*

Proof. Given a valid membership word $w_0, \dots, w_{N-1}$, extend it periodically. By [theorem 2.2](#), it avoids every forbidden pattern. Its unique semantic flags produce a periodic six-state walk, and quotienting gives a closed four-state walk. Start immediately before reading vertex 0, so there is no choice of a rotation or of a distinguished vertex.

Conversely, a closed four-state labeled walk lifts uniquely by [equation \(13\)](#). Its sequence of emitted bits is periodic, and the lifted states satisfy [equations \(10\) and \(11\)](#). The word cannot be all ones, so [theorem 3.1](#) identifies these flags with the semantic flags. Hence all forbidden patterns are absent. By [theorems 2.1 and 2.2](#), the corresponding subset is mutually visible.

These constructions are inverse: the membership word fixes the semantic flags, then all six states, and hence the four states. Each selected vertex emits exactly one label 1, so weights are preserved. $\square$

The (i, i) entry of $M(y)^N$ enumerates the closed walks with initial state i . Summing over initial states gives the trace. This proves [equation \(5\)](#) for $n \geq 2$. For $n = 1$, $\text{tr } M(y)^2 = 1 + 2y$, while $V_1(y) = 1 + 2y + y^2$, proving the remaining case.

Remark 3.3 (Why there is no factor $1/N$). A subset is a word on fixed vertices, not a necklace. Its state at the boundary before vertex 0 is forced. Trace sums the possible states at that fixed boundary; it does not freely choose a boundary along the word. Dividing the trace by N would therefore be incorrect. Even periodic words have one anchored state sequence, not several copies arising from a symmetry.

4 Generating functions and recurrences

4.1 The full cardinality generating function

The characteristic polynomial of $M(y)$ is

$$F(z, y) = z^4 - z^3 - yz^2 - y(1 + y)z + y(1 + y). \quad (14)$$

Define $\mathcal{V}(x, y) = \sum_{n \geq 1} V_n(y)x^n$. From the trace theorem and a formal matrix geometric series,

$$\begin{aligned} \mathcal{V}(x, y) &= y^2x + x \operatorname{tr} \left(M(y)^2 \sum_{j \geq 0} x^j M(y)^{3j} \right) \\ &= y^2x + x \operatorname{tr} (M(y)^2 (I - xM(y)^3)^{-1}). \end{aligned} \quad (15)$$

The inverse exists as a formal power series because its matrix argument has constant term I .

Theorem 4.1 (Bivariate rational form). *The generating function is $\mathcal{V}(x, y) = P(x, y)/D(x, y)$, where*

$$\begin{aligned} D(x, y) = & 1 - (3y^2 + 6y + 1)x \\ & + y(3y^3 + 11y^2 + 12y + 3)x^2 \\ & - y^2(y + 1)^3(y + 3)x^3 + y^3(y + 1)^3x^4, \end{aligned} \quad (16)$$

$$\begin{aligned} P(x, y) = & (1 + y)^2x - y(3y^3 + 7y^2 + 6y + 3)x^2 \\ & + y^2(3y^4 + 10y^3 + 11y^2 + 6y + 3)x^3 \\ & - y^3(y + 1)^3(y^2 + 3y + 1)x^4 + y^5(y + 1)^3x^5. \end{aligned} \quad (17)$$

Proof. Starting from equation (15), multiply by $D = \det(I - xM^3)$ and use the adjugate formula. Direct expansion gives equation (16) and

$$P = y^2xD + x \operatorname{tr}(M^2 \operatorname{adj}(I - xM^3)),$$

whose expansion is equation (17). These are finite polynomial identities, independently checked by the supplied symbolic certificate. $\square$

Putting $y = 1$ in equations (16) and (17) yields precisely equation (3), completing the proof of theorem 1.1. As a basic check on cardinality marking, at $y = 0$ the rational expression reduces to $x/(1 - x)$: there is one empty set for each graph.

4.2 The recurrence and its initial exception

Corollary 4.2. *The sequence a_n satisfies*

$$a_n = 10a_{n-1} - 29a_{n-2} + 32a_{n-3} - 8a_{n-4} \quad (n \geq 6), \quad (18)$$

with initial values

$$a_1 = 4, \quad a_2 = 21, \quad a_3 = 127, \quad a_4 = 749, \quad a_5 = 4455.$$

The eventual recurrence has minimal order four over $\mathbb{Q}$.

Proof. Compare coefficients in equation (3). The numerator has degree five, so the homogeneous relation begins at $n = 6$. The numerator and denominator are coprime over $\mathbb{Q}$; the reduced denominator has degree four, which is the minimum order of any eventual constant-coefficient recurrence. $\square$

Starting equation (18) at $n = 5$ would give 4447, not 4455. The residual is 8, exactly the coefficient of x^5 in the numerator. Alternatively set $c_1 = 3$ and $c_n = a_n$ for $n \geq 2$. Then $c_n = \operatorname{tr} M(1)^{3n-1}$ satisfies the same homogeneous recurrence for $n \geq 5$. The change at $n = 1$ accounts for all of the discrepancy.

The coefficient polynomials themselves satisfy, for $n \geq 6$,

$$\begin{aligned} V_n = & (3y^2 + 6y + 1)V_{n-1} - y(3y^3 + 11y^2 + 12y + 3)V_{n-2} \\ & + y^2(y + 1)^3(y + 3)V_{n-3} - y^3(y + 1)^3V_{n-4}. \end{aligned} \quad (19)$$

The first polynomials are

$$\begin{aligned}
 V_1 &= 1 + 2y + y^2, \\
 V_2 &= 1 + 5y + 10y^2 + 5y^3, \\
 V_3 &= 1 + 8y + 28y^2 + 48y^3 + 34y^4 + 8y^5, \\
 V_4 &= 1 + 11y + 55y^2 + 154y^3 + 242y^4 + 198y^5 + 77y^6 + 11y^7, \\
 V_5 &= 1 + 14y + 91y^2 + 350y^3 + 847y^4 + 1274y^5 \\
 &\quad + 1141y^6 + 576y^7 + 147y^8 + 14y^9.
 \end{aligned}$$

The coefficients of $1, y, y^2$ are $1, N, \binom{N}{2}$, as required directly by the definition. The appendix records further scalar values, while the archive supplies the entire coefficient triangle through $n = 50$.

5 Return blocks and a finite binomial formula

5.1 First returns to the state B

The automaton has only two types of first-return walk from B to B :

$$B \xrightarrow{0} A \xrightarrow{0} \dots \xrightarrow{0} A \xrightarrow{1} B, \quad B \xrightarrow{1} C \xrightarrow{0} D \xrightarrow{\varepsilon} B, \quad \varepsilon \in \{0, 1\}. \quad (20)$$

The first type has $j \geq 0$ self-loops at A , length $j + 2$, and weight y . The second has length three and weights y and y^2 . Thus, if z marks walk length, the first-return series is

$$H(z, y) = \frac{yz^2}{1-z} + y(1+y)z^3. \quad (21)$$

Every closed walk other than the all- A loop visits B . A walk that never visits A consists entirely of B, C, D rounds and has length divisible by three. Our lengths $N = 3n - 1$ are not divisible by three, so every nonzero membership word under consideration has at least one first-type block.

There is a useful scalar determinant identity:

$$\det(I - zM(y)) = (1 - z)(1 - H(z, y)). \quad (22)$$

For example, expansion of the right side gives $1 - z - yz^2 - y(1+y)z^3 + y(1+y)z^4$, the reciprocal characteristic polynomial from [equation \(14\)](#).

If $T_N(y) = \text{tr } M(y)^N$, logarithmic differentiation gives the formal identity

$$\sum_{N \geq 1} T_N(y) z^N = -z \frac{d}{dz} \log \det(I - zM(y)) = \frac{z}{1-z} + \frac{zH_z(z, y)}{1-H(z, y)}. \quad (23)$$

One can verify the first equality without any spectral assumptions: Jacobi's determinant formula gives $z \text{tr}((I - zM)^{-1}M) = \sum_{N \geq 1} \text{tr}(M^N)z^N$.

Expanding $-\log(1 - H)$ in [equation \(23\)](#) yields

$$T_N(y) = 1 + N[z^N] \sum_{r \geq 1} \frac{H(z, y)^r}{r}. \quad (24)$$

This identity automatically handles periodic arrangements of blocks. In particular, the denominator r is not an unsupported division of a necklace count by its length.

5.2 An explicit count of every cardinality

Theorem 5.1 (Binomial formula). *Let $n \geq 2$ and $N = 3n - 1$. Then*

$$V_n(y) = 1 + \sum_{\substack{p \geq 1, q \geq 0 \\ 2p+3q \leq N}} \frac{N}{p} \binom{p+q-1}{q} \binom{N-p-3q-1}{p-1} y^{p+q} (1+y)^q. \quad (25)$$

In particular, with the convention that $\binom{a}{b} = 0$ outside $0 \leq b \leq a$,

$$\begin{aligned} [y^k] V_n(y) &= \mathbf{1}_{\{k=0\}} \\ &+ \sum_{\substack{p \geq 1, q \geq 0 \\ 2p+3q \leq N}} \frac{N}{p} \binom{p+q-1}{q} \binom{N-p-3q-1}{p-1} \binom{q}{k-p-q}. \end{aligned} \quad (26)$$

Proof. In H^{p+q} choose p first-type blocks and q second-type blocks. The coefficient contribution before extracting z is

$$\frac{N}{p+q} \binom{p+q}{p} y^{p+q} (1+y)^q z^{2p+3q} (1-z)^{-p}.$$

If $j = N - 2p - 3q \geq 0$, then $[z^j](1-z)^{-p} = \binom{j+p-1}{p-1}$. Also

$$\frac{1}{p+q} \binom{p+q}{p} = \frac{1}{p} \binom{p+q-1}{q}.$$

Terms with $p = 0$ have length divisible by three and do not contribute. Substitution into equation (24) proves equation (25); expanding $(1+y)^q$ gives equation (26). $\square$

Although individual displayed summands can be rational because of the factor N/p , the complete coefficients are integers by the proved closed-walk interpretation. The implementation uses exact rational arithmetic for this formula, independently of the integer transfer-matrix algorithm.

6 Maximum sets and fixed deficits from the maximum

6.1 A block defect identity

Take a nonempty valid word and decompose its closed walk into first-return blocks. Let $p \geq 1$ count first-type blocks, q count second-type blocks, j count all A self-loops, and d count second-type blocks whose final label is 0 rather than 1. If K is the number of selected vertices, then

$$N = 2p + j + 3q, \quad K = p + 2q - d, \quad 2N - 3K = p + 2j + 3d. \quad (27)$$

The right side is a nonnegative weighted defect with $p \geq 1$. Unlike a loose density bound, this identity retains exactly the information needed to count near-extremal sets.

Theorem 6.1 (Maximum mutual visibility). *For $n \geq 2$,*

$$\max_{S \in \mathcal{M}_n} |S| = 2n - 1, \quad [y^{2n-1}] V_n(y) = 3n - 1. \quad (28)$$

The maximum sets are exactly the cyclic shifts of the binary word

$$01(101)^{n-1}. \quad (29)$$

Proof. From equation (27), $3K \leq 2N - 1 = 6n - 3$, so $K \leq 2n - 1$. For equality, $p + 2j + 3d = 1$, forcing $p = 1$ and $j = d = 0$. The length equation then gives $q = n - 1$. Conversely, these parameters define a valid closed walk: one block $B \rightarrow A \rightarrow B$ and $n - 1$ blocks $B \rightarrow C \rightarrow D \rightarrow B$, all with last label 1. Its labels are equation (29).

There is exactly one visit to state A . Placing that state at any of the N anchored positions determines the entire walk, giving N different walks. The bijection of theorem 3.2 makes their words different as well. Thus all N cyclic shifts are distinct and exhaust the maximum sets. $\square$

The exceptional graph G_1 has maximum cardinality two and a unique maximum set. It is not covered by theorem 6.1.

Corollary 6.2 (One below maximum). *For $n \geq 2$,*

$$[y^{2n-2}]V_n(y) = (3n - 1) \left(2(n - 1) + \frac{1}{4} \binom{n}{3} \right). \quad (30)$$

Proof. The defect is now four, so the possibilities for (p, j, d) are $(4, 0, 0)$, $(2, 1, 0)$, and $(1, 0, 1)$. Their respective contributions, from the block formula, are

$$\frac{N}{4} \binom{n}{3}, \quad N(n - 1), \quad N(n - 1).$$

When $n = 2$, the first case is impossible and its binomial coefficient is zero. This proves the formula throughout the stated range. $\square$

6.2 Every fixed deficit gives an eventual polynomial

Define

$$E_h(n) = [y^{2n-1-h}]V_n(y), \quad h \geq 0. \quad (31)$$

The case $h = 0$ is the maximum count. For fixed h , the possible block defects form a finite set independent of n .

Theorem 6.3 (Fixed-deficit polynomiality). *Fix an integer $h \geq 0$. For every $n \geq \max(2, 2h + 1)$,*

$$E_h(n) = \sum_{\substack{p \geq 1, j, d \geq 0 \\ p+2j+3d=1+3h}} \frac{3n-1}{p} \binom{p+q-1}{p-1} \binom{j+p-1}{p-1} \binom{q}{d}, \quad q = n - \frac{1+2p+j}{3}. \quad (32)$$

In particular, $E_h(n)$ is eventually a polynomial in n of degree $3h + 1$, with leading term

$$E_h(n) = \frac{3}{(3h+1)!} n^{3h+1} + O(n^{3h}) \quad (n \rightarrow \infty; h \text{ fixed}). \quad (33)$$

Proof. For cardinality $K = 2n - 1 - h$, the left side of the defect identity is $2N - 3K = 1 + 3h$. Thus the nonnegative parameters satisfy the constraint in equation (32). Solving the length equation gives the displayed q . It is an integer because $p + 2j \equiv 1 \pmod{3}$ implies $1 + 2p + j \equiv 0 \pmod{3}$.

There are $\binom{j+p-1}{p-1}$ ways to distribute the j self-loops among p first-type blocks, and $\binom{q}{d}$ ways to choose the low-weight second-type blocks. The same cyclic block factor as in equation (25) is $(3n - 1) \binom{p+q-1}{p-1} / p$. This proves the sum whenever the terms with $q < d$ are omitted.

To obtain one uniform polynomial range, write $c = (1 + 2p + j)/3$. The constraint gives

$$c + d = 1 + 2h - j - d \leq 1 + 2h.$$

Consequently $n \geq 2h + 1$ implies $q = n - c \geq d$ for every allowed triple. No summands need to be omitted in that range.

For a fixed triple, q is affine in n . Interpreting the binomial coefficients as their falling-factorial polynomials, the degree of its summand is

$$1 + (p - 1) + d = p + d = 1 + 3h - 2j - 2d \leq 3h + 1.$$

Equality holds only when $j = d = 0$, in which case $p = 3h + 1$. This unique highest-degree summand has leading coefficient

$$\frac{3}{p} \cdot \frac{1}{(p - 1)!} = \frac{3}{(3h + 1)!}.$$

There can therefore be no cancellation of the leading term, proving both the degree and equation (33). $\square$

For example,

$$\begin{aligned} E_0(n) &= 3n - 1, \\ E_1(n) &= \frac{(n - 1)(3n - 1)(n^2 - 2n + 48)}{24}, \end{aligned} \tag{34}$$

$$\begin{aligned} E_2(n) &= \frac{3n - 1}{5040} (n^6 - 9n^5 + 445n^4 - 2535n^3 \\ &\quad + 17194n^2 - 40296n + 30240), \quad n \geq 5. \end{aligned} \tag{35}$$

The bound in [theorem 6.3](#) is sufficient, not claimed to be optimal. For $h = 1$, [theorem 6.2](#) proves the polynomial one index earlier than the uniform theorem guarantees. The supplied symbolic script derives the polynomials for $h = 0, 1, 2, 3, 4$ by exact expansion of [equation \(32\)](#), not by numerical interpolation.

7 Asymptotic enumeration and the size of a random set

7.1 Exponential growth with an explicit error bound

For each real $y > 0$, the matrix $M(y)$ is nonnegative and its directed graph is strongly connected. The loop at A makes it aperiodic, hence primitive. Let $\lambda(y)$ be its Perron eigenvalue. It is positive, simple, and strictly larger in modulus than every other eigenvalue.

Write $\lambda = \lambda(1)$. From [equation \(14\)](#), it is the root greater than one of

$$z^4 - z^3 - z^2 - 2z + 2 = 0. \tag{36}$$

That root is unique: for $z > 1$, the equation is equivalent to

$$z = \frac{1}{z - 1} + \frac{2}{z^2}, \tag{37}$$

whose left side increases and right side decreases. It is indeed a Perron root, since $((z - 1)^{-1}, 1, 2z^{-2}, 2z^{-1})^T$ is a strictly positive eigenvector. Evaluating [equation \(36\)](#) at the rational numbers 1.82 and 1.83 isolates it to (1.82, 1.83).

Let ρ_2, ρ_3, ρ_4 be the other roots, counted with multiplicity, and set $\eta = \max_{i \geq 2} |\rho_i| < \lambda$. The trace identity gives

$$a_n = \lambda^{3n-1} + \rho_2^{3n-1} + \rho_3^{3n-1} + \rho_4^{3n-1} \quad (n \geq 2). \tag{38}$$

Therefore

$$|a_n - \lambda^{3n-1}| \leq 3\eta^{3n-1}. \quad (39)$$

This bound is exact when λ and η are defined algebraically as above; the decimals below are only evaluations of those constants.

Corollary 7.1 (Asymptotic count). *Let $\alpha = \lambda^3$ and $C = \lambda^{-1}$. Then*

$$\begin{aligned} a_n &\sim C\alpha^n, \\ \alpha &= 6.039002874613972596691496575757 \dots, \\ C &= 0.549133899324504062872716935976 \dots \end{aligned} \quad (40)$$

The constant α is the largest positive root of $t^4 - 10t^3 + 29t^2 - 32t + 8$.

Numerically,

$$\begin{aligned} \lambda &= 1.821049476694320862024323558251 \dots, \\ \rho_2 &= 0.702029547174037673333882475778 \dots, \\ \rho_{3,4} &= -0.761539511934179267679103017015 \dots \pm 0.992207523189564457635687644537 \dots i, \\ \eta &= 1.250767043582024533276068104631 \dots \end{aligned}$$

The conjugate pair explains the small oscillatory correction to pure exponential growth. No fitted growth parameter is used: all of these constants come from the proved transfer matrix.

7.2 The probability generating function

Choose S uniformly from all a_n mutual-visibility sets, including the empty set, and let $K_n = |S|$. Then

$$\mathbb{E}[y^{K_n}] = \frac{V_n(y)}{V_n(1)}. \quad (41)$$

Because the Perron eigenvalue is simple, $\lambda(y)$ is analytic in a complex neighborhood of $y = 1$. Shrinking that neighborhood if necessary, the other eigenvalues are uniformly separated in modulus from it. The trace of their powers is exponentially smaller, so uniformly near $s = 0$,

$$\mathbb{E}[e^{sK_n}] = \left(\frac{\lambda(e^s)}{\lambda(1)} \right)^N (1 + O(\theta^N)), \quad N = 3n - 1, \quad (42)$$

for some $0 < \theta < 1$. The amplitude of the leading trace term is exactly one. This is why there is no extra constant-order correction when the mean is written in terms of N .

Define

$$L(s) = \log \lambda(e^s), \quad \mu = L'(0), \quad \sigma^2 = L''(0). \quad (43)$$

Here σ^2 denotes the variance *per vertex*, not the variance of K_n itself.

Theorem 7.2 (Mean, variance, and a central limit theorem). *There exists $\kappa \in (0, 1)$ such that*

$$\mathbb{E}K_n = \mu(3n - 1) + O(\kappa^{3n-1}), \quad \text{Var } K_n = \sigma^2(3n - 1) + O(\kappa^{3n-1}), \quad (44)$$

where

$$\mu = \frac{\lambda^2 + 3\lambda - 3}{\lambda(4\lambda^3 - 3\lambda^2 - 2\lambda - 2)} = 0.370526040195751563090487009118 \dots, \quad (45)$$

$$\sigma^2 = \frac{46\lambda^3 + 18\lambda^2 - 40\lambda + 4}{490\lambda^3 - 57\lambda^2 - 468\lambda + 166} = 0.128918101249168837027097668809 \dots > 0. \quad (46)$$

Moreover,

$$\frac{K_n - \mu(3n - 1)}{\sqrt{\sigma^2(3n - 1)}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (47)$$

Proof. Differentiating the analytic logarithm of equation (42) at zero gives equation (44). Uniform analyticity controls the derivatives of the error by Cauchy's estimates; one may enlarge the exponential rate from θ to a number $\kappa < 1$ to absorb any polynomial factors.

For the constants, implicit differentiation of $F(\lambda(y), y) = 0$ gives

$$\lambda'(y) = -\frac{F_y}{F_z}, \quad \lambda''(y) = -\frac{F_{zz}(\lambda')^2 + 2F_{zy}\lambda' + F_{yy}}{F_z}, \quad (48)$$

with all partial derivatives evaluated at $(\lambda(y), y)$. At $y = 1$,

$$F_y = -\lambda^2 - 3\lambda + 3, \quad F_z = 4\lambda^3 - 3\lambda^2 - 2\lambda - 2,$$

so $\mu = \lambda'(1)/\lambda$ is equation (45). The chain rule gives

$$\sigma^2 = \frac{\lambda'(1) + \lambda''(1)}{\lambda} - \left(\frac{\lambda'(1)}{\lambda} \right)^2. \quad (49)$$

Substituting equation (48) and reducing with $\lambda^4 = \lambda^3 + \lambda^2 + 2\lambda - 2$ yields equation (46). Its numerator and denominator are both positive throughout $1.82 < \lambda < 1.83$; direct rational lower bounds follow by using 1.82 in positive terms and 1.83 in negative terms. Thus positivity is not inferred merely from the decimal approximation.

For completeness, fix $t \in \mathbb{R}$ and set $s = it/\sqrt{\sigma^2 N}$. Taylor expansion gives

$$L(s) - L(0) = \mu s + \frac{1}{2}\sigma^2 s^2 + O(s^3).$$

Multiply by N , subtract the centering term μNs , and use equation (42). The characteristic function of the standardized variable tends to $e^{-t^2/2}$, since $N|s|^3 \rightarrow 0$ and the spectral error tends to zero. The continuity theorem for characteristic functions proves equation (47). $\square$

Thus a uniformly chosen mutual-visibility set contains asymptotically about 37.05% of the vertices, whereas the largest possible sets contain almost 66.67%. The near-maximum polynomial counts and the central Gaussian regime describe different parts of the same enumerator; neither is being used as an approximation for the other.

8 Exact algorithms and reproducibility

8.1 Linear-time recognition

A direct implementation of the visibility definition examines many pairs and shortest paths. For this graph family, theorem 3.1 gives a much smaller recognizer. Given $n \geq 2$ and a membership word of length N : choose a zero z , set $b_z = 0$, and visit the remaining residues by adding three modulo N . At each visited position t , compute $b_t = w_t(w_{t-1} \vee b_{t-3})$. Finally reject exactly when some $b_t w_{t+1} = 1$. The all-ones word is rejected separately, and every word is accepted for $n = 1$.

There are N flag computations and N final checks. This is $O(N)$ Boolean work and $O(N)$ auxiliary bits in the straightforward implementation. It does not construct the graph's $\Theta(N^2)$ adjacency relation. The code validates the input length and the binary alphabet.

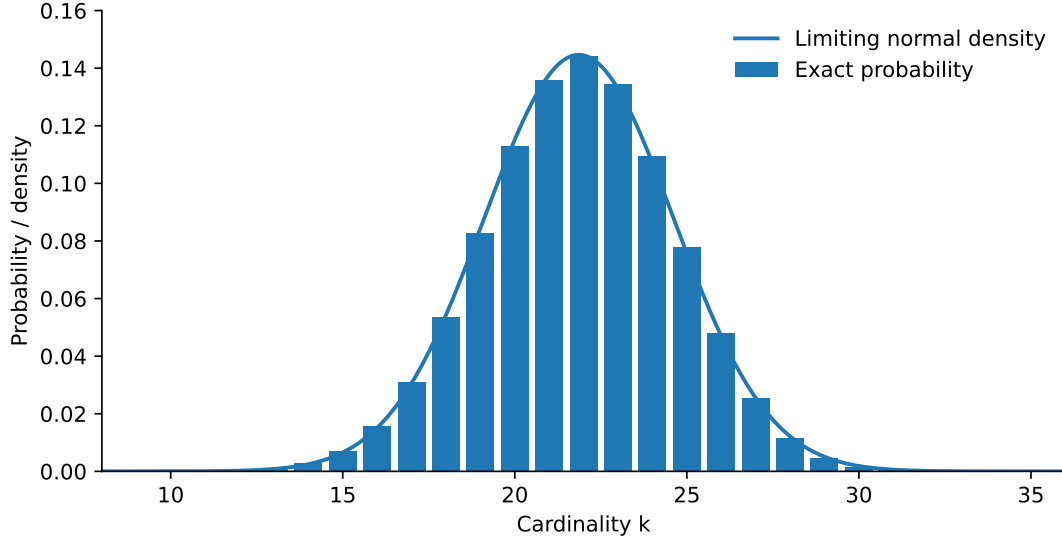


Figure 2: Exact cardinality probabilities for $n = 20$ ($N = 59$), calculated from $V_{20}(y)$, with the normal density using the limiting mean and variance from [theorem 7.2](#). The line is an illustration, not an additional claim of a local central limit theorem or an error bound at this finite n .

8.2 Integer counting and coefficient computation

To count all sets, repeated squaring computes $\text{tr } M(1)^{3n-1}$ with $O(\log n)$ multiplications of fixed-size matrices of integers. Alternatively, [equation \(18\)](#) generates the first m terms in $O(m)$ integer arithmetic operations with constant working storage, apart from the output. These are arithmetic-operation bounds: the integers have $\Theta(n)$ bits, so neither statement asserts constant-cost machine arithmetic.

The polynomial implementation propagates labeled edges, maintaining for each initial and current state an array indexed by the number of ones. This computes $V_n(y)$ in $O(N^2)$ additions of nonnegative integers and $O(N)$ stored coefficients. It is independent of the rational coefficients appearing in [equation \(25\)](#). The supplied second implementation evaluates that binomial formula using exact fractions and checks agreement.

8.3 Exact uniform sampling

A transfer matrix can do more than enumerate. Let $A = M(1)$ and precompute A^r for $0 \leq r \leq N$. Choose a starting state s with probability

$$\frac{(A^N)_{ss}}{\text{tr } A^N}. \quad (50)$$

With current state u and $r + 1$ edges remaining, choose a particular labeled edge $e : u \rightarrow v$ with probability

$$\frac{(A^r)_{vs}}{(A^{r+1})_{us}}. \quad (51)$$

The two $D \rightarrow B$ edges must remain separate choices. Their multiplicity is already counted in the denominator. Emit each chosen edge's bit.

For any particular closed labeled walk, the successive completion counts telescope. Multiplying [equations \(50\) and \(51\)](#) gives probability $1/\text{tr } A^N$. By the anchored bijection, the emitted

subset is therefore exactly uniform over $\mathcal{M}_n$. For $n = 1$, simply choose both bits independently with probability one half.

The implementation draws integer random numbers below the exact total weight, not floating-point approximations to probabilities. Preprocessing and each sample use $O(N)$ operations on large integers. The cached powers occupy $O(N^2)$ bits, since a fixed-size matrix at exponent r uses $O(r)$ bits per entry. Random-bit generation and integer arithmetic costs are not hidden behind a claim of constant bit complexity.

8.4 Independent tests actually performed

The tests were run locally for this report. Their machine-readable output is included in the archive. They use three different descriptions of visibility, as well as independently evaluated algebraic consequences.

Check	Range	Observed result
Direct common-neighbor graph predicate versus Boolean flag recognizer	Every subset for $1 \leq n \leq 8$; 9,586,980 subsets	No mismatches; every coefficient agrees with the transfer enumerator.
Shortest-path BFS definition versus flag recognizer	Every subset for $1 \leq n \leq 5$; 18,724 subsets	No mismatches; no diameter-two assumption used by the BFS predicate.
Actual labeled closed walks versus accepted membership masks	Lengths $3n - 1$, $1 \leq n \leq 5$	Every expected mask appears exactly once; only the stated $n = 1$ full-set exception.
Recurrence versus integer matrix powers	$1 \leq n \leq 250$	Exact agreement; first 15 terms also match OEIS.
Binomial formula versus polynomial dynamic programming	$2 \leq n \leq 40$	All coefficients agree exactly.
Fixed-deficit formula versus polynomial coefficients	$2 \leq n \leq 40$, $0 \leq h \leq \min(9, 2n - 2)$	Exact agreement.
Symbolic matrix and rational identities	Characteristic polynomial, resolvent numerator and denominator, variance formula	Exact polynomial/rational identities verified with SymPy.
Uniform sampler validity smoke test	7,000 words across $n = 1, 2, 3, 5, 8, 20, 50$	Every sampled word passes recognition. Uniformity follows from the telescoping proof, not this test.

The C++ checker constructs adjacency directly from [equation \(1\)](#). For each nonedge $\{u, v\}$ it forms $\{u, v\} \cup (\Gamma(u) \cap \Gamma(v))$ and tests whether that obstruction is contained in the subset. Separately, it constructs the cyclic pattern obstructions and checks equality of the two families. It then compares the graph predicate against the flag recognizer for *every* subset.

The independent Python BFS check is closer still to the original definition. For a selected source, it allows a search to reach another selected vertex but forbids expanding that vertex. A target is visible precisely when its restricted distance equals its distance in the original graph. Thus it does not invoke the common-neighbor lemma or use the transfer matrix to decide visibility.

The symbolic checker verifies the adjugate identity used in [theorem 4.1](#), not merely the first several coefficients of a guessed series. It also derives the fixed-deficit polynomials by finite symbolic sums. None of these checks substitutes for the all- n combinatorial proof.

8.5 Files and execution

The main executable module uses only the Python 3.9 or later standard library. SymPy is needed only for the symbolic certificate, and Matplotlib only to regenerate the distribution figure. A C++17 compiler is needed for the larger exhaustive comparison. From the extracted directory, the central commands are:

```
python code/verify.py
python code/symbolic_certificate.py
python code/visibility.py --n 20 --sample
python code/visibility.py --write-data data

g++ -O3 -std=c++17 code/exhaustive.cpp -o exhaustive
./exhaustive 8 > data/exhaustive.json
python code/verify.py

pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The README gives the corresponding workflow and explains the outputs. A SHA-256 manifest covers the deliverable files. No compiler binary, virtual environment, font file, or fetched third-party paper is required in the ZIP.

9 What the argument establishes, and what it does not

The selected conjecture is resolved by a bijection, followed by a finite matrix calculation. Its rational denominator is the characteristic data of $M(y)^3$, because the graph order advances by three when n increases by one. The nonproper numerator records the genuine exceptional first graph; it is not a numerical nuisance to be suppressed.

The strongest structural refinement is that the entire cardinality polynomial comes from the same four-state system. The return-block decomposition then gives a finite formula for every coefficient, exact maximum-set classification, and an infinite family of eventual polynomial formulas indexed by the deficit h . Perron theory controls typical sets, while the defect identity controls the extreme tail.

Several matters are deliberately left outside the claims. We do not assert that the four-state presentation is the unique or minimal deterministic presentation of the underlying binary language. The minimality statement proved above concerns the scalar eventual recurrence, not automata. We do not claim that the size polynomials have real zeros or any unproved global log-concavity property. A central limit theorem is proved, but no local limit theorem or finite- n normal-approximation error is asserted. Finally, the literature search documents a conjecture that remained labeled open in the consulted source; it does not establish exclusive priority for every corollary derived here.

The method suggests concrete further investigations without requiring any of them for the present result: enumeration modulo graph automorphisms, additional vertex weights, and related circulant families whose common-neighbor obstructions admit finite-state compression. Such extensions must check periodic-state multiplicities anew. The coprimality and zero-reset arguments in this proof are precisely the features that should not be silently assumed to survive a change of graph family.

A Sequence values and coefficient data

The first fifteen entries below agree with the values displayed by OEIS on the consultation date [1]. The remaining displayed entries and the extended file are calculated from the proved recurrence and independently checked by matrix powers.

n	a_n	n	a_n
1	4	11	213,940,607
2	21	12	1,291,991,757
3	127	13	7,802,356,615
4	749	14	47,118,492,485
5	4,455	15	284,548,794,383
6	26,725	16	1,718,391,139,389
7	161,007	17	10,377,369,263,383
8	971,613	18	62,668,963,071,925
9	5,866,487	19	378,458,048,186,527
10	35,426,517	20	2,285,509,240,092,589

For illustration, the next two size polynomials after those in [section 4](#) are

$$\begin{aligned}
 V_6(y) &= 1 + 17y + 136y^2 + 663y^3 + 2142y^4 + 4692y^5 \\
 &\quad + 6885y^6 + 6579y^7 + 3944y^8 + 1394y^9 + 255y^{10} + 17y^{11}, \\
 V_7(y) &= 1 + 20y + 190y^2 + 1120y^3 + 4505y^4 + 12844y^5 \\
 &\quad + 26190y^6 + 37820y^7 + 37910y^8 + 25680y^9 \\
 &\quad + 11292y^{10} + 3000y^{11} + 415y^{12} + 20y^{13}.
 \end{aligned}$$

The file `data/visibility_coefficients.csv` has columns `n`, `vertices`, `cardinality`, `count`. It records nonzero coefficient ranges, including the constant coefficient, through $n = 50$. The file `data/b391632_extended.txt` has 250 indexed terms in b-file-style format. These are locally derived data; the file is not represented as an official OEIS submission.

B A dependency map for auditing the proof

The logical dependencies can be audited without running any code:

1. The circulant definition gives the exact common-neighbor set in [theorem 2.1](#). This step is where graph geometry enters.
2. A selected nonedge is blocked precisely by selecting all common neighbors, giving [equation \(8\)](#). The periodic-extension lemma removes the graph-dependent upper bound on pattern length.
3. The flags remember every possible finite prefix by [equation \(10\)](#). The zero-reset argument and $\gcd(3, 3n - 1) = 1$ prove that periodic flags cannot introduce spurious multiplicity or unsupported states.
4. The six semantic states exhaust the valid possibilities. The explicit incoming-edge lift proves that their four-state quotient preserves closed labeled walks.
5. The anchored bijection justifies the trace identity. Everything thereafter follows from matrix algebra, first-return blocks, or the Perron eigenvalue.

The main edge cases are the empty word, the all-ones word, the full subset of G_1 , patterns crossing vertex 0, patterns longer than one period, and repeated cyclic words. Each is handled explicitly in sections 2 and 3. The fixed-deficit theorem excludes the empty-set target by its stated eventual range; no zero-cardinality term has been inadvertently included in its block sum.

References

- [1] E. W. Weisstein and A. Howroyd, *A391632: Number of mutual-visibility sets in the n -Andrásfai graph*, The On-Line Encyclopedia of Integer Sequences. Entry created 14 December 2025; generating function conjectured by A. Howroyd, 12 January 2026; consulted 19 September 2026. <https://oeis.org/A391632>.
- [2] G. Di Stefano, *Mutual Visibility in Graphs*, arXiv:2105.02722, version 2, 15 July 2021. <https://arxiv.org/abs/2105.02722>.
- [3] A. Behtoei, S. Payrovi, and S. B. Pejman, *Metric dimension of Andrásfai graphs*, arXiv:1706.06852, 21 June 2017. The circulant definition is in the introduction. <https://arxiv.org/abs/1706.06852>.